Drawing a desktop from scratch

A presentation editor in Odin, on a software rasterizer

palmy · slides.bin

01

Why a slide editor

It is the hardest thing you can ask of a 2D renderer

The slide is the model

- A deck is a title, a theme and a list of slides
- A slide is a layout, speaker notes and elements in paint order
- An element is a rectangle in a fixed 1280x720 space
- Colours name theme slots, so one keystroke restyles the deck
- The whole thing is plain data, so JSON is the native format

One renderer, four jobs

- The editing canvas
- The sorter thumbnails
- The fullscreen projector
- The PNG exporter

- Scaled from slide space
- Measured at the size it draws
- Same text wrapping everywhere
- A thumbnail is the real slide

Pictures composite like anything else



stb_image decodes PNG, JPEG, BMP, GIF, TGA and more into straight 0xAARRGGBB pixels. The sampler here is bilinear, rounds the corners and follows the element's rotation.

Shapes, strokes, rotation

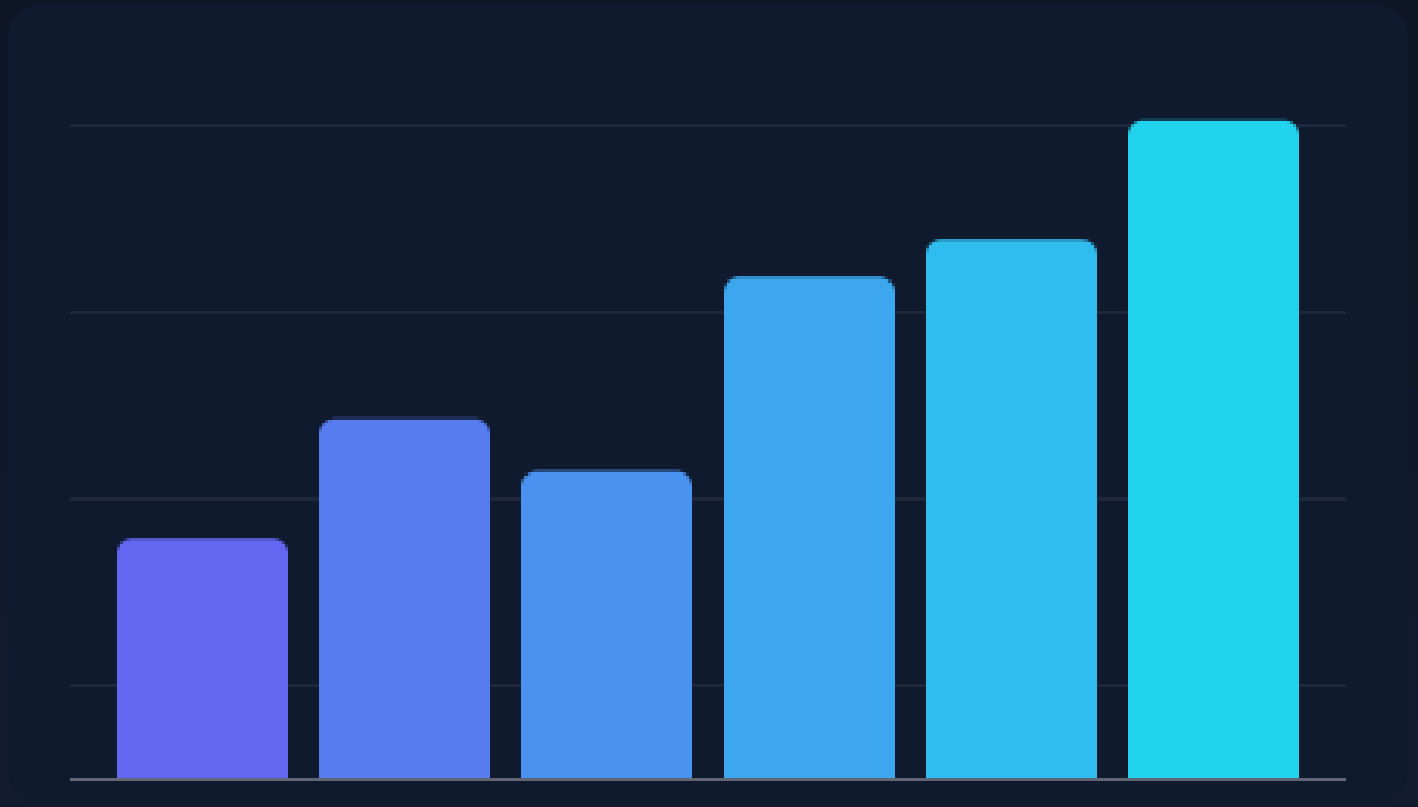


Every edge is a signed distance field, evaluated in the shape's own rotated frame.

What it costs

3

dynamic libraries:
libc, libm, vdso



Frames drawn per second, 1080p, one core

Press Escape to leave the projector

F5 presents · Tab is the presenter view · Ctrl+E exports PNGs